

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2023-24

Paper No. STB-301: Probability Theory and Statistical Methods

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory)

- (a) Distinguish between standard deviation and standard error. If $X_i \sim N(\mu, \sigma^2)$ for $i = 1, 2, \dots, n$, find the standard error of the sample mean $\bar{X}$ and the sample variance S^2 . [4]
- (b) What do you mean by sample proportion? Derive the sampling distribution of the sample proportion and obtain its standard error. [10]

Question 2.

- (a) What are the different criteria of a good estimator? Explain them briefly. [6]
- (b) If $X_1, X_2, \dots, X_n$ is a random sample from a normal population $N(\mu, \sigma^2)$, show that the sample variance $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased and consistent estimator for the parameter σ^2 . [8]

Question 3.

- (a) What do you mean by simple and composite hypothesis? A random sample of size $n = 1$ is taken from an exponential distribution with parameter λ (pdf $f(x) = \lambda e^{-\lambda x}, x > 0$). If $x \geq 1$ is the critical region for testing the hypothesis $H_0 : \lambda = 2$ against $H_1 : \lambda = 1$, find the type-I error, type-II error and the power of the test. [8]
- (b) Define χ^2 (chi-square) distribution, derive the expression of its probability density function, and show that the skewness of the χ^2 distribution with n degrees of freedom is $\sqrt{8/n}$. [6]

Question 4.

- (a) Define Student's t -distribution. Obtain the even order moments of the t -distribution and discuss the shape of the distribution. [6]
- (b) Define Fisher's F -distribution and show that if $t \sim t(n)$, then $t^2 \sim F(1, n)$. [4]
- (c) Describe the χ^2 goodness-of-fit test for testing independence of attributes. [4]

Question 5.

- (a) What do you mean by paired t -test? State the test of significance for the paired t -test....
[6]
- (b) Explain the testing procedure for testing the equality of two variances from two independent normal populations. [8]
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Question 6.

- (a) Describe the importance of the Central Limit Theorem. State the test of significance for a single proportion in the case of a large sample. Also compute the probable limits for the population proportion. [8]
- (b) What is Fisher's Z -transformation? Describe any application of this transformation. [6]
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Question 7.

- (a) Define order statistics along with its various applications. Obtain the probability distribution of the r -th order statistics $X_{(r)}$. [6]
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from a continuous uniform distribution $U(0, \theta)$. Obtain the probability density function of the smallest and largest order statistics. Also compute the mean and variance of these distributions. [8]
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Question 8. Explain the various advantages and disadvantages of non-parametric tests. Describe the sign test and run test in detail. [14]